SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA0312	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
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Section / Batch:	2024	**Date:**	22.10.226

Code of Conduct

I P.Ranjithkumar (192411119)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Ranjithkumar

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Mobile IP Tunnel Overhead

Question: A logistics vehicle sends a 500-byte data packet. The Mobile IP tunnel adds 20 bytes of overhead per packet. Calculate the additional overhead introduced by the tunnel.

Given: Data packet = 500 bytes; tunnel overhead = 20 bytes per packet.

Formula: Additional overhead = overhead added to each packet

Solution:

- The tunnel adds 20 bytes to every packet.
- For one packet: overhead = 20 × 1 = 20 bytes.
- If useful, total transmitted size = 500 + 20 = 520 bytes.

Final Answer: 20 bytes of additional overhead per packet (total packet size becomes 520 bytes).

Note: The overhead is 20 bytes, not 20% of the packet. Relative to the original 500-byte payload, it is (20/500) × 100 = 4%.

2. Maximum Theoretical Data Rate

Question: A wireless network has 20 MHz bandwidth and uses 64-QAM, which carries 6 bits per symbol. Estimate the maximum theoretical data rate.

Given: Bandwidth = 20 MHz = 20×10^6 Hz; bits per symbol = 6.

Formula: Assuming one symbol per hertz per second: Data rate = bandwidth × bits per symbol

Solution:

- Convert bandwidth: 20 MHz = 20,000,000 symbols/s under the idealized assumption.
- Multiply by 6 bits per symbol: 20,000,000 × 6 = 120,000,000 bits/s.
- Convert to Mbps: 120,000,000 ÷ 1,000,000 = 120 Mbps.

Final Answer: 120 Mbps (idealized theoretical estimate).

Note: This estimate assumes a symbol rate equal to the 20 MHz bandwidth and ignores coding, guard intervals, pilots, and protocol overhead. Actual throughput is lower.

3. Routing Table Memory Requirement

Question: Each mobile node maintains 50 routing-table entries, and each entry uses 32 bytes. Calculate the total memory required.

Given: Number of entries = 50; size per entry = 32 bytes.

Formula: Total memory = number of entries × size per entry

Solution:

- Substitute the values: 50 × 32 bytes.
- Calculate: 50 × 32 = 1,600 bytes.
- In decimal kilobytes: 1,600 ÷ 1,000 = 1.6 kB (approximately 1.56 KiB using 1 KiB = 1,024 bytes).

Final Answer: 1,600 bytes (about 1.6 kB or 1.56 KiB).

4. Distance Travelled Between Routing Updates

Question: An emergency vehicle travels at a constant speed of 10 m/s. The routing protocol updates information every 2 seconds. Find the distance travelled between updates.

Given: Speed = 10 m/s; time interval = 2 s.

Formula: Distance = speed × time

Solution:

- Substitute: distance = 10 m/s × 2 s.
- Multiply: distance = 20 m.

Final Answer: 20 metres between consecutive routing updates.

5. DSR Route Discovery

Question: During DSR route discovery, a route request passes through 5 intermediate nodes before reaching the destination. Each node forwards the request only once. Find the number of route request transmissions.

Given: Intermediate nodes = 5; each forwards once.

Formula: Under the stated linear-path assumption: transmissions = intermediate-node forwards + source transmission

Solution:

- The source sends the initial route request: 1 transmission.
- Each of the 5 intermediate nodes forwards it once: 5 transmissions.
- Total transmissions up to the destination = 1 + 5 = 6.

Final Answer: 6 route-request transmissions, assuming a single path with 5 intermediate nodes.

Note: If the question counts only requests forwarded by the 5 intermediate nodes and excludes the source's initial transmission, the count is 5. In a real broadcast network, duplicate receptions and multiple paths can change the count.

6. Mobile IP Subnet Addressing

Question: A subnet uses the mask 255.255.255.240. Calculate the maximum number of usable host addresses in that subnet.

Given: Subnet mask = 255.255.255.240 (/28).

Formula: Host bits = 32 − prefix length; usable hosts = 2^(host bits) − 2

Solution:

- The mask 255.255.255.240 is /28, so host bits = 32 − 28 = 4.
- Total addresses = $2^4 = 16$.
- For a conventional IPv4 subnet, subtract the network and broadcast addresses: 16 − 2 = 14.

Final Answer: 14 usable IPv4 host addresses.

Note: This uses conventional IPv4 subnet rules; special-purpose point-to-point configurations may use different allocation rules.

7. Spectral Efficiency

Question: A wireless system has channel bandwidth 20 MHz and symbol rate 1 Msps. Calculate spectral efficiency using the given symbol rate.

Given: Bandwidth = 20 MHz; symbol rate = 1 Msymbol/s.

Formula: Symbol-rate-to-bandwidth ratio = symbol rate ÷ bandwidth

Solution:

- Convert the values to the same scale: 1 Msps = 1×10^6 symbols/s; 20 MHz = 20×10^6 Hz.
- Ratio = $(1 \times 10^6) \div (20 \times 10^6) = 0.05$ symbol/s/Hz.
- This is a symbol-rate efficiency ratio. To calculate bit/s/Hz, the number of bits per symbol (modulation order) is also needed.

Final Answer: 0.05 symbol/s/Hz as a symbol-rate-to-bandwidth ratio.

Note: True spectral efficiency is normally expressed in bit/s/Hz. Since modulation bits per symbol are not provided, the bit/s/Hz value cannot be uniquely determined. If each symbol carried b bits, efficiency would be 0.05b bit/s/Hz.

8. Network Density

Question: A wireless sensor network contains 50 nodes over a deployment area of 1 km². Each node has a 250 m transmission range. Calculate the approximate node density.

Given: Number of nodes = 50; area = 1 km².

Formula: Node density = number of nodes ÷ deployment area

Solution:

- Density = 50 ÷ 1 km².
- Therefore, density = 50 nodes/km².

- Since 1 km^2 = 1,000,000 m^2, density = 50 ÷ 1,000,000 = 0.00005 nodes/m^2.

Final Answer: 50 nodes/km^2 (or 0.00005 nodes/m^2).

Note: The transmission range does not affect basic nodes-per-area density. If the intended meaning is coverage or neighborhood density, node placement and overlap information would also be needed.

9. Doppler Shift

Question: A train moves at 30 m/s and the wireless carrier frequency is 900 MHz. Calculate the maximum magnitude of the Doppler shift.

Given: Speed = 30 m/s; carrier frequency = 900 MHz = 900×10^6 Hz; speed of light $c \approx 3 \times 10^8$ m/s.

Formula: Maximum Doppler shift magnitude: $f_D = (v/c) \times f_c$

Solution:

- Substitute: $f_D = (30 \div (3 \times 10^8)) \times (900 \times 10^6)$.
- Compute $30 \div (3 \times 10^8) = 1 \times 10^{-7}$.
- Then $f_D = 1 \times 10^{-7} \times 900 \times 10^6 = 90$ Hz.

Final Answer: Maximum Doppler shift magnitude = 90 Hz.

Note: The shift is positive when moving toward the transmitter and negative when moving away; the exact value depends on the direction of motion relative to the signal.

10. Packet Transmission Time

Question: A WLAN has a data rate of 54 Mbps. A packet contains 1,500 bytes. Calculate the time needed to transmit the packet, excluding other delays.

Given: Data rate = 54 Mbps = 54×10^6 bits/s; packet size = 1,500 bytes.

Formula: Transmission time = packet size in bits ÷ data rate

Solution:

- Convert bytes to bits: 1,500 × 8 = 12,000 bits.
- Transmission time = 12,000 ÷ (54×10^6) seconds.
- Time ≈ 0.0002222 s. Convert to milliseconds: 0.0002222 × 1,000 ≈ 0.222 ms.

Final Answer: Approximately 0.000222 seconds, or 0.222 milliseconds (222

microseconds).

Note: This is the packet's serialization/transmission time only. It excludes contention, preamble, headers not included in the 1,500 bytes, propagation delay, and retransmissions.